Resolution Method

in Propositional Logic

Thanh-Sach LE

Faculty of Computer Science and Engineering
HoChiMinh City University of Technology

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Outline

1. What is Hilbert System?
2. Axioms
3. Inference Rule
4. Examples

What is Hilbert System?

Definition

A Hilbert system is a deductive system

Hilbert system consists of:

1. A set of wffs called axioms
2. A set of inference rules. An inference rule consists of (i) a list of wffs called the premises of the rule, (ii) a wff called the rule's conclusion. Usually, inference rules do not have more than two premises.

What is Hilbert System?

Hilbert System's Application

Input:

1. A set of premises.
2. A conclusion.

Application:

- Use Hilbert System to prove the conclusion from the given premises.

Axiom 1

$\alpha \Rightarrow (\beta \Rightarrow \alpha)$

Why?

α	β	$\beta \Rightarrow \alpha$	$\alpha \Rightarrow (\beta \Rightarrow \alpha)$
F	F	T	**T**
F	T	F	**T**
T	F	T	**T**
T	T	T	**T**

$\alpha \Rightarrow (\beta \Rightarrow \alpha)$ is true for any α and β.
$\alpha \Rightarrow (\beta \Rightarrow \alpha)$ is a tautology.

Axiom 2

$$[\alpha \Rightarrow (\beta \Rightarrow \gamma)] \Rightarrow [(\alpha \Rightarrow \beta) \Rightarrow (\alpha \Rightarrow \gamma)]$$

Why?

$s \triangleq [\alpha \Rightarrow (\beta \Rightarrow \gamma)]$

$t \triangleq [(\alpha \Rightarrow \beta) \Rightarrow (\alpha \Rightarrow \gamma)]$

α	β	γ	$\alpha \Rightarrow \beta$	$\alpha \Rightarrow \gamma$	$\beta \Rightarrow \gamma$	s	t	**Axiom 2**
F	F	F	T	T	T	T	T	**T**
F	F	T	T	T	T	T	T	**T**
F	T	F	T	T	F	T	T	**T**
F	T	T	T	T	T	T	T	**T**
T	F	F	F	F	T	T	T	**T**
T	F	T	F	T	T	T	T	**T**
T	T	F	T	F	F	F	F	**T**
T	T	T	T	T	T	T	T	**T**

Axiom 3

$$(\neg\alpha \Rightarrow \neg\beta) \Rightarrow (\beta \Rightarrow \alpha)$$

Why?

α	β	$\neg\alpha$	$\neg\beta$	$\neg\alpha \Rightarrow \neg\beta$	$\beta \Rightarrow \alpha$	**Axiom 3**
F	F	T	T	T	T	**T**
F	T	T	F	F	F	**T**
T	F	F	T	T	T	**T**
T	T	F	F	T	T	**T**

Axiom 3 is true for any α and β.
Axiom 3 is a tautology.

Axiom 1

$\alpha \Rightarrow (\beta \Rightarrow \alpha)$

Axiom 2

$[\alpha \Rightarrow (\beta \Rightarrow \gamma)] \Rightarrow [(\alpha \Rightarrow \beta) \Rightarrow (\alpha \Rightarrow \gamma)]$

Axiom 3

$(\neg\alpha \Rightarrow \neg\beta) \Rightarrow (\beta \Rightarrow \alpha)$

Modus ponens (MP)

$$\frac{\alpha, \alpha \Rightarrow \beta}{\beta}$$

Examples

Problem

Premises:

1. Nothing

Conclusion:

- $A \Rightarrow A$

Problem

Premises:

1. Nothing

Conclusion:

- $A \Rightarrow A$

No.	Formulas	Notes
1.	$(A \Rightarrow ((A \Rightarrow A) \Rightarrow A))$	**Axiom 1**, with
	$\alpha = A, \beta = (A \Rightarrow A), \gamma = A$	
2.	$(A \Rightarrow ((A \Rightarrow A) \Rightarrow A)) \Rightarrow (A \Rightarrow (A \Rightarrow A)) \Rightarrow (A \Rightarrow A))$	**Axiom 2**, with
	$\alpha = A, \beta = (A \Rightarrow A), \gamma = A$	
3.	$(A \Rightarrow (A \Rightarrow A)) \Rightarrow (A \Rightarrow A))$	**1, 2, MP**
4.	$A \Rightarrow (A \Rightarrow A))$	**Axiom 1**, with
	$\alpha = A, \beta = A, \gamma = A$	
5.	$(A \Rightarrow A)$	**3, 4, MP**
	■	**Proved**

Problem

Premises:

1. $A \Rightarrow B$
2. $B \Rightarrow C$

Conclusion:

- $A \Rightarrow C$

No.	Formulas	Notes
1.	$(A \Rightarrow B)$	**Premise**
2.	$(B \Rightarrow C)$	**Premise**
3.	$((B \Rightarrow C) \Rightarrow (A \Rightarrow (B \Rightarrow C)))$	**Axiom 1, with**
	$\alpha = (B \Rightarrow C), \beta = A$	
4.	$(A \Rightarrow (B \Rightarrow C))$	**2, 3, MP**
5.	$(A \Rightarrow (B \Rightarrow C)) \Rightarrow ((A \Rightarrow B) \Rightarrow (A \Rightarrow C))$	**Axiom 2, with**
	$\alpha = A, \beta = B, \gamma = C$	
6.	$((A \Rightarrow B) \Rightarrow (A \Rightarrow C))$	**4, 5,MP**
7.	$(A \Rightarrow C)$	**1, 6, MP**
	$\blacksquare$	**Proved**